

Normal ligand binding and signaling by CD47 (integrin-associated protein) requires a long range disulfide bond between the extracellular and membrane-spanning domains.

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CD47 is a unique member of the Ig superfamily with a single extracellular Ig domain followed by a multiply membrane-spanning (MMS) domain with five transmembrane segments, implicated in both integrin-dependent and -independent signaling cascades. Essentially all functions of CD47 require both the Ig and MMS domains, raising the possibility that interaction between the two domains is required for normal function. Conservation of Cys residues among CD47 homologues suggested the existence of a disulfide bond between the Ig and MMS domains that was confirmed by chemical digestion and mapped to Cys(33) and Cys(263). Subtle changes in CD47 conformation in the absence of the disulfide were suggested by decreased binding of two anti-Ig domain monoclonal antibodies, decreased SIRPalpha1 binding, and reduced CD47/SIRPalpha1-mediated cell adhesion. Mutagenesis to prevent formation of this disulfide completely disrupted CD47 signaling independent of effects on ligand binding, as assessed by T cell interleukin-2 secretion and Ca(2+) responses. Loss of the disulfide did not affect membrane raft localization of CD47 or its association with alpha(v)beta(3) integrin. Thus, a disulfide bond between the Ig and MMS domains of CD47 is required for normal ligand binding and signal transduction.

Deficit of CD47 results in a defect of marginal zone dendritic cells, blunted immune response to particulate antigen and impairment of skin dendritic cell migration.

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CD47 is a ubiquitously expressed cell surface glycoprotein that associates with integrins and regulates chemotaxis, migration, and activation of leukocytes. CD47 is also a ligand for signal regulatory protein alpha, a cell surface receptor expressed on monocytes, macrophages, granulocytes, and dendritic cell (DC) subsets that regulates cell activation, adhesion, and migration. Although the function of CD47 in macrophages and granulocytes has been studied in detail, little is known about the role of CD47 in DC biology in vivo. In this study we demonstrate that CD47(-/-) mice exhibit a selective reduction of splenic CD11c(high)CD11b(high)CD8alpha(-)CD4(+) DCs. These DCs correspond to marginal zone DCs and express signal regulatory protein alpha, possibly explaining their selective deficiency in CD47(-/-) mice. Deficiency of marginal zone DCs resulted in impairment of IgG responses to corpusculate T cell-independent Ags. Although epidermal DCs were present in normal numbers in CD47(-/-) mice, their migration to draining lymph nodes in response to contact sensitization was impaired, while their maturation was intact. In vitro, CD47(-/-) mature DCs showed normal CCR7 expression but impaired migration to CCL-19, whereas immature DC response to CCL-5 was only slightly impaired. These results demonstrate a fundamental role of CD47 in DC migration in vivo and in vitro and in the function of marginal zone DCs.

Preclinical and Clinical Development of Therapeutic Antibodies Targeting Functions of CD47 in the Tumor Microenvironment.

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CD47 is a ubiquitously expressed cell surface glycoprotein that functions as a signaling receptor for

thrombospondin-1 and as the counter-receptor for signal regulatory protein- α (SIRP α). Engaging SIRP α on macrophages inhibits phagocytosis, and CD47 thereby serves as a physiological marker of self. However, elevated CD47 expression on some cancer cells also protects tumors from innate immune surveillance and limits adaptive antitumor immunity via inhibitory SIRP α signaling in antigen presenting cells. CD47 also mediates inhibitory thrombospondin-1 signaling in vascular cells, T cells, and NK cells, and blocking inhibitory CD47 signaling on cytotoxic T cells directly increases tumor cell killing. Therefore, CD47 functions as an innate and adaptive immune checkpoint. These findings have led to the development of antibodies and other therapeutic approaches to block CD47 functions in the tumor microenvironment. Preclinical studies in mice demonstrated that blocking CD47 can limit the growth of hematologic malignancies and solid tumors and enhance the efficacy of conventional chemotherapy, radiation therapy, and some targeted cancer therapies. Humanized CD47 antibodies are showing promise in early clinical trials, but side effects related to enhanced phagocytic clearance of circulating blood cells remain a concern. Approaches to circumvent these include antibody preloading strategies, development of antibodies that recognize tumor-specific epitopes of CD47, SIRP α antibodies, and bivalent antibodies that restrict CD47 blockade to specific tumor cells. Preclinical and clinical development of antibodies and related biologics that inhibit CD47/SIRP α signaling are reviewed, including strategies to combine these agents with various conventional and targeted therapeutics to improve patient outcome for various cancers.

The development of small-molecule inhibitors targeting CD47.

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Immunotherapy has become an indispensable part of cancer treatment. A pivotal phagocytosis checkpoint, named cluster of differentiation 47 (CD47), which functions as 'don't eat me' signal to protect cells from phagocytosis upon interaction with signal regulatory protein alpha (SIRP α) on macrophages, has recently attracted much attention. Numerous antibodies targeting the CD47/SIRP α axis have shown encouraging efficacy in clinical trials. Meanwhile, studies on small-molecule inhibitors that interfere with CD47/SIRP α interaction or regulate CD47 expression are also in full swing. In this review, we summarize the small-molecule inhibitors interrupting the binding of CD47/SIRP α and regulating CD47 at the transcriptional, translational, and post-translational modification (PTM) levels. We provide perspectives and strategies for targeting the CD47/SIRP α phagocytosis checkpoint.

A CD47-associated super-enhancer links pro-inflammatory signalling to CD47 upregulation in breast cancer.

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CD47 is a cell surface molecule that inhibits phagocytosis of cells that express it by binding to its receptor, SIRP α , on macrophages and other immune cells. CD47 is expressed at different levels by neoplastic and normal cells. Here, to reveal mechanisms by which different neoplastic cells generate this dominant 'don't eat me' signal, we analyse the CD47 regulatory genomic landscape. We identify two distinct super-enhancers (SEs) associated with CD47 in certain cancer cell types. We show that a set of active constituent enhancers, located within the two CD47 SEs, regulate CD47 expression in different cancer cell types and that disruption of CD47 SEs reduces CD47 gene expression. Finally we report that

the TNF-NFKB1 signalling pathway directly regulates CD47 by interacting with a constituent enhancer located within a CD47-associated SE specific to breast cancer. These results suggest that cancers can evolve SE to drive CD47 overexpression to escape immune surveillance.

Is CD47 a potentially promising therapeutic target in cardiovascular diseases? - Role of CD47 in cardiovascular diseases.

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CD47 (cluster of differentiation 47) is a ubiquitously expressed transmembrane protein that belongs to the immunoglobulin superfamily. CD47 is both a receptor for the matricellular protein thrombospondin-1 (TSP-1) and a ligand for signal-regulatory protein alpha (SIRP α). Suppression of CD47 activity enhances angiogenesis and blood flow, restores phagocytosis by macrophages, improves ischemic tissue survival, attenuates ischemia reperfusion injury, and reverses atherosclerotic plaque formation. In conclusion, these observations suggest a pathogenic role of CD47 in the development of cardiovascular diseases (CVDs) and indicate that CD47 might be a potentially promising molecular target for treating CVDs. Herein, we highlight the role of CD47 in the CVD pathogenesis and discuss the potential clinical application by targeting CD47 for treating CVDs.

CD47 mediates the progression of colorectal cancer by inducing tumor cell apoptosis and angiogenesis.

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CD47 is an immunoregulatory protein that is found on the cell surface and plays significant roles in cellular functions such as proliferation, apoptosis, migration, and immune homeostasis. CD47 is overexpressed in various human cancers and is associated with tumor development, progression, and poor prognosis. In this study, we analyzed the expression of CD47 to determine whether it affected the oncogenic behavior of colorectal cancer (CRC) and investigated the prognostic value of CD47 expression in patients with CRC. We investigated 468 patients with CRC who underwent curative surgery and examined the expression of CD47 in tumor and lymph node tissues by performing RT-PCR and immunohistochemistry. Apoptosis, proliferation, angiogenesis, and lymphangiogenesis were determined via a TUNEL assay and immunohistochemical staining for Ki-67, CD34, and D2-40. CD47 expression was increased in human CRC tumors and metastatic lymph nodes compared with normal colorectal mucosa and non-metastatic lymph node tissues. CD47 expression was significantly associated with perineural invasion, lymphovascular invasion, cell differentiation, cancer stage, depth of invasion, lymph node metastasis, distant metastasis, and poor survival. The mean apoptotic index and microvessel density value of CD47-positive tumors were significantly higher than those of CD47-negative tumors. However, no significant difference was observed between CD47 expression and Ki-67 labeling index or lymphatic vessel density. These results indicate that CD47 mediated the progression of CRC by inducing tumor cell apoptosis and angiogenesis.

An antitumor peptide RS17-targeted CD47, design, synthesis, and antitumor activity.

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CD47 is a widely expressed transmembrane protein located on the surface of somatic cells. It mediates a variety of cellular processes including apoptosis, proliferation, adhesion, and migration. An important role for CD47 is the transmission of a "Don't eat me" signal by interacting with SIRP α on the macrophage surface membrane, thereby preventing the phagocytosis of normal cells. However, cancer cells can take advantage of this autogenous signal to protect themselves from phagocytosis, thus enabling immune escape. Blocking the interaction between CD47 and SIRP α has proven to be effective in removing cancer cells. The treatment of various cancers with CD47 monoclonal antibodies has also been validated. We designed and synthesized a peptide (RS17), which can specifically bind to CD47 and block CD47-SIRP α signaling. The affinity of RS17 for CD47-expressing tumor cells was determined, while the inhibition of CD47-SIRP α signaling was evaluated in vitro and in vivo. The results indicated that RS17 significantly promotes the phagocytosis of tumor cells by macrophages and had a similar therapeutic effect compared with a positive control (CD47 monoclonal antibodies). In addition, a cancer xenograft mouse model was established using CD47-expressing HepG2 cells to evaluate the effect of RS17 on tumor growth in vivo. Using ex vivo and in vivo mouse models, RS17 demonstrated a high inhibitory effect on tumor growth. Based on our results, RS17 may represent a novel therapeutic peptide for cancer therapy.

Targeting the cluster of differentiation 47/signal-regulatory protein alpha axis in myeloid malignancies.

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The antitumor activity of macrophages is regulated by a balance of prophagocytic and antiphagocytic signals. Cluster of differentiation 47 (CD47), the dominant macrophage immune checkpoint ('do not eat me' signal), interacts with its receptor signal-regulatory protein alpha (SIRP α) to suppress phagocytic activities. This axis plays a pivotal role in immune evasion in myeloid malignancies as well as multiple cancers providing strong rationale for therapeutic exploitation. Preclinical studies have revealed overexpression of CD47 on leukemic stem cells and myeloblasts from patients with myelodysplastic syndromes (MDS) and acute myeloid leukemia (AML), which contributes to immune surveillance evasion and is associated with poor outcomes. Blockade of CD47 with different approaches has demonstrated proof-of-concept antitumor activities mainly through phagocytic clearance. Early phase clinical trials combining the anti-CD47 mAb magrolimab with the hypomethylating agent azacitidine have showed synergistic activities, deep and durable responses, as well as a tolerable safety profile in these patients, including those with TP53 mutations. Targeting CD47/SIRP α axis, in combination with other therapeutic agents, represents a promising treatment approach for patients with myeloid malignancies, particularly the challenging TP53-mutated subgroup.

Characterization of cluster of differentiation 47 expression and its potential as a therapeutic target in esophageal squamous cell cancer.

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The increased expression of cluster of differentiation (CD)47 has been identified in a number of different tumor types and is recognized as an adverse prognostic factor that indicates an increased risk of mortality in patients. The binding of CD47 to signal regulatory protein α (SIRP α) inhibits the macrophage

phagocytosis of tumor cells by triggering an inhibitory 'do not eat me' signal. This is one of the mechanisms used by tumor cells to evade immune surveillance. In the present study, CD47 levels and macrophage infiltration were assessed in patients with esophageal squamous cell cancer (ESCC). CD47-overexpressing ESCC cell lines were selected and human M2 macrophage phagocytic activity was measured. The results revealed that CD47 is highly expressed and macrophages are markedly infiltrated in cancerous tissue compared with non-cancerous tissue. High CD47 expression was detected in ESCC cell lines and the results of a phagocytosis assay indicated that human M2 macrophages phagocytized tumor cells in a dose-dependent manner following the blocking of CD47-SIRP α signaling by anti-CD47 antibodies. The results of the present study therefore support the use of anti-CD47 immunotherapy to treat patients with ESCC.
